Adam Raman

Product Strategy | AI Innovation | Built Environment Technology

Sendai, Japan · Malaysian · English (native) · Malay (native) · Japanese (professional working proficiency)

Innovation strategist and former founder with **12+ years bridging physical infrastructure and digital solutions**. I founded and ran a design-build practice for ten years — growing revenue roughly twofold each year for eight consecutive years, profitably, and cutting project costs 20–30% through process redesign, early BIM-based workflows and new contract formats.

Since 2020 I have applied automation and AI to real problems: **a voice-controlled Zigbee / Google Home smart home, AI-assisted research and calculation with every output verified by me, and — from 2025 — AI-assisted software deployed to production**, including a retrieval-grounded advisor, a tools directory and a bilingual book. I am currently a Building Energy & Automation Engineer in Japan, building internal automation and AI documentation tools, with doctoral research on solar-regenerated passive dehumidification.

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EXPERIENCE	AI PRACTICE	FOUNDER RECORD	BASED IN
12+ years across architecture, product & AI	5 years — home automation, AI-assisted research, AI-built software	40+ design-build projects, full P&L	Sendai, Japan — open worldwide

2020 — PRESENT

AI & Technology Practice

Applied automation and AI built alongside professional roles and research — in stages, from a voice-controlled home to software deployed in production. This is the core of my current direction: product and innovation roles where AI has to work in the real world.

2020 **Home automation & IoT**

A **voice-controlled home** by 2021, with most functions linked through Google Home over Zigbee and Wi-Fi — first field-tested on client projects at Lakar, then built out in my own house as a live test bed.

2022 **AI as a research partner — human in the loop**

From the launch of ChatGPT in late 2022, and later other assistants, I used AI to **research, draft and refine papers and calculations** — with me checking and verifying every output against the sources and the maths.

2025 **AI-assisted software, taken to finished products**

From 2025, AI-assisted coding from idea to finished product: **Demon Hunter**, a browser game with a custom Web Audio engine (built locally in 2025, deployed online in 2026); and the bilingual **Momotaro** children's book — HTML-coded layout, AI illustration — **published on Amazon KDP in December 2025**.

2026 **Retrieval-grounded AI & LLM workflows**

Online Sheikh: a multi-step Gemini workflow that extracts the question's keywords, retrieves verified Quran and Hadith passages from public APIs, and grounds its answer in what it retrieved. Plus **AIBO**, a Gemini-powered receptionist that greets and guides visitors on my portfolio site.

2026 **Process automation from standards**

HydroCalc digitises Japanese Industrial Standards (JIS) pipe-flow charts into a live calculator for water-supply network design. Further JIS-based calculators followed, built with AI after studying the standard chapter by chapter in NotebookLM.

PhD **Research data & modelling, AI-assisted**

For my doctoral research: **sensor-data pipelines** that decode proprietary logger files into clean, interoperable datasets, and **hygrothermal models** (sorption isotherm, combined temperature and ambient-moisture weighting) of a solar-regenerated desiccant test box — built with AI assistance and checked against measured data.

Deployed, not just coded locally

Online Sheikh — islamic-advisor.pages.dev	Cloudflare Pages + Functions · Gemini
Portfolio site + AIBO receptionist — solar-punk-five.vercel.app	Next.js · Vercel · Gemini
ADAMTOOL — adamtool.pages.dev	Cloudflare Pages · Supabase
PDF → DXF converter	Python backend on Render
HydroCalc — github.com/lakar-team/hydrocalc	JIS pipe-flow calculator · GitHub

THE THREAD THROUGH IT ALL

Taking messy real-world data and processes, finding the underlying pattern, and teaching AI to handle them in defined, repeatable steps. That pattern runs through my design work, my teaching, and my applied AI builds — it is how I approach product problems.

CAPABILITIES

Skills

A portable mix: deep technical fluency in AI and the built environment, plus the product, commercial and leadership skills that come from running a practice.

AI & DATA

Applied AI & Engineering

Retrieval-grounded AI apps · LLM workflows · AI-assisted development · Data structuring · Process automation · Python / TypeScript / Next.js · Simulation & modelling

Retrieval-grounded Gemini apps, LLM workflows, sensor-data preparation and standards-based calculators — shipped to production on Vercel and Cloudflare.

PRODUCT & STRATEGY

Product & Leadership

Human-centric design | Design thinking | Product-market fit | Business development | P&L management | Stakeholder management | Mentorship

Ten years running a practice with full P&L responsibility — finding fit, redesigning processes, and leading multidisciplinary teams and stakeholders.

BUILT ENVIRONMENT TECH

Domain Technology

BIM workflows — early adopter | Revit (basic) | AutoCAD — advanced | Rhino / Grasshopper (PhD lab) | IoT & smart buildings | Energy systems design

Early BIM adoption, parametric design, IoT smart-home integration, and energy systems — the domain depth behind the technology work.

CAREER

Experience

From architecture practice to founding a design-build company, to research and applied AI in Japan. The roles change; the work — redesign the process, build the system — stays the same.

Dec 2025 — Present
Sendai, Japan

Building Energy & Automation Engineer

ReFiL G.K.

Building internal automation and AI tools inside a working design practice.

- Builds calculators from Japanese Industrial Standards (JIS) for benchmarking and regulatory compliance
- Builds AI-assisted documentation workflows
- Energy audits and mechanical / electrical system proposals

2022 — Present
Student 2022–2025; on leave since 2025, researching part-time
Sendai, Japan

Doctoral Researcher

Tohoku University

Doctoral research on a solar-regenerated charcoal desiccant system for passive dehumidification in hot, humid climates — cooling energy load reductions of up to 50% measured under specific hot, humid conditions. Experiments and hygrothermal modelling.

- Teaching Assistant (English as a second language), 2023–2025

2012 — 2022
Malaysia

Founder, Managing Director & Design Lead

Lakar Design

Founded and ran a design-build practice for ten years with full P&L responsibility across delivery, clients and commercial growth.

- 40+ design-build projects delivered (residential, commercial fit-outs, smart homes)
- BIM-based 3D workflow adopted across the practice; in 2012, helped build the 3D model of the Malaysian MRT system, ahead of government BIM adoption
- Smart-home integration (Google Home, Zigbee) field-tested on client projects, then in my own home
- Grew annual revenue from ~RM10k (2012) to ~RM1.5M, roughly doubling for eight consecutive years, profitable throughout
- Process redesign, BIM workflow and new contract formats cut project costs 20–30%

8 years
Revenue roughly doubling yearly, all profitable

20–30%
Project costs cut via process, BIM and contracts

40+
Design-build projects, full P&L

2016 — 2019
Malaysia

Assistant Lecturer & Facilitator

Universiti Teknologi MARA (UiTM)

- Design thinking curriculum development and delivery
- Mentored 200+ students

2009 & 2011–2012
Malaysia (MArch in the UK between)

Assistant Architect → Architect

S&A Architects

Assistant architect in 2009; returned as architect in 2011 after completing my MArch, and left in 2012 to found Lakar Design. The low-rise housing master plan I designed in 2011 (Denai Alam, Phase J15) won a PAM Award for innovation in design in 2017, once construction was complete.

BACKGROUND

Education & Credentials

MArch (Part 2) — 2011

University of Manchester

Dissertation: “Barriers to Energy Efficiency in Existing Housing Stock” — the starting point for a career-long focus on energy, buildings and systems.

BSc Architecture (Part 1) — 2008

Universiti Teknologi MARA (UiTM)

Dean’s Award, top of class.

Qualifications ARB/RIBA Part 2 (UK) · LAM Part 2 (Malaysia)

English	Native (C2)
Malay	Native
Japanese	Professional working proficiency (JLPT N2–N3 equivalent)
Nationality	Malaysian
Location	Sendai, Japan — open to relocation / hybrid / remote

GET IN TOUCH

Contact

I’m interested in product strategy, AI innovation and built-environment technology roles — especially where messy operational reality needs to become a clear, repeatable system. The fastest way to reach me is email or LinkedIn.

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